CHAPTER 6

SPECIAL TYPE SEMICONDUCTOR DEVICES

Thiruvalluvar reitrated the importance of light in the following couplets

ஒளிஒருவற்கு உள்ள வெறுக்கை இளிஒருவற்கு
அஃதிறந்து வாழ்தும் எனல். # குறள் 971

விளக்கம்
ஒருவனுக்கு ஒளி ஊக்கமிகுதியே ஆகும், ஒருவனுக்கு இழிவு அந்த ஊக்கம் இல்லாமலேயே உயிர்வாழலாம் என்று எண்ணுதலாம்.

English Couplet 971:

The light of life is mental energy; disgrace is his
Who says, 'I 'ill lead a happy life devoid of this.'

Couplet Explanation:
One's light is the abundance of one's courage; one's darkness is the desire to live destitute of such (a state of mind).

CONTENT

6.1 Light Emitting Diode (LED)
6.2. Seven Segment LED
6.3. Light Dependent Resistor (LDR)
6.4. Liquid Crystal Display (LCD)
6.5. Photo Diode
6.6. Photo Transistor
6.7. Solar Cell
6.8. Unipolar Junction Transistor (UJT)
6.9. Field Effect Transistor (FET)
6.10. Metal Oxide Semiconductor Field Effect Transistor (MOSFET)
6.11. Silicon Controlled Rectifier (SCR)
6.12. DIAC
6.13. TRIAC
6.14. Insulated Gate Bipolar Transistor (IGBT)
6.15 Integrated Circuit (IC)

LEARNING OBJECTIVES

While learning this Chapter, the student can

1. **Understand the basic concepts of special type of semiconductor devices.**
2. **Describe the working principles of special semiconductor devices.**
3. **Understand the switching action of Thyristor devices.**
4. **Know about various applications of the semiconductor devices.**

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